Exploratory Data Analysis Report

Tropical Cyclone Dataset with Vedic-Astronomical Features

47,533 observations • 944 storms • 186 features

1. Overview and Data Description

The dataset contains 47,533 6-hourly best-track observations from 944 tropical cyclones (2016–2026), enriched with 186 features across three groups:

• Meteorological (14): latitude, longitude, wind speed, spline track geometry, elevation, land proximity.

• Vedic-Panchang (18): Tithi, Nakshatra, Yoga, Karana, Paksha, Moonsign, Samvatsara, Dinamana/Ratrimana, Weekday, Ayana, Ritu.

• Scientific-Astronomical (154): planet positions, altitudes, azimuths, angular separations, Vedic degree representations.

Wind speed (kt) is the target variable. Median = 40 kt; mean ≈ 45 kt; right-skewed (skewness = 1.35). All distributions are non-normal (Shapiro–Wilk confirmed), so non-parametric tests are used throughout.

2. Feature Importance: Tree-Model Ensemble

Three tree-based models (Random Forest, XGBoost, LightGBM) were trained with 5-fold GroupKFold cross-validation (groups = storm_id) to prevent within-storm autocorrelation leakage. Feature importances were averaged across folds and models.

• longitude: mean importance = 0.0588

• latitude: mean importance = 0.0420

• spline_end_lon: mean importance = 0.0340

• dinamana_minutes: mean importance = 0.0207

• vedic_chandra_speed_deg_per_day: mean importance = 0.0203

• Vedic features in top-30: 10 of 30 positions.

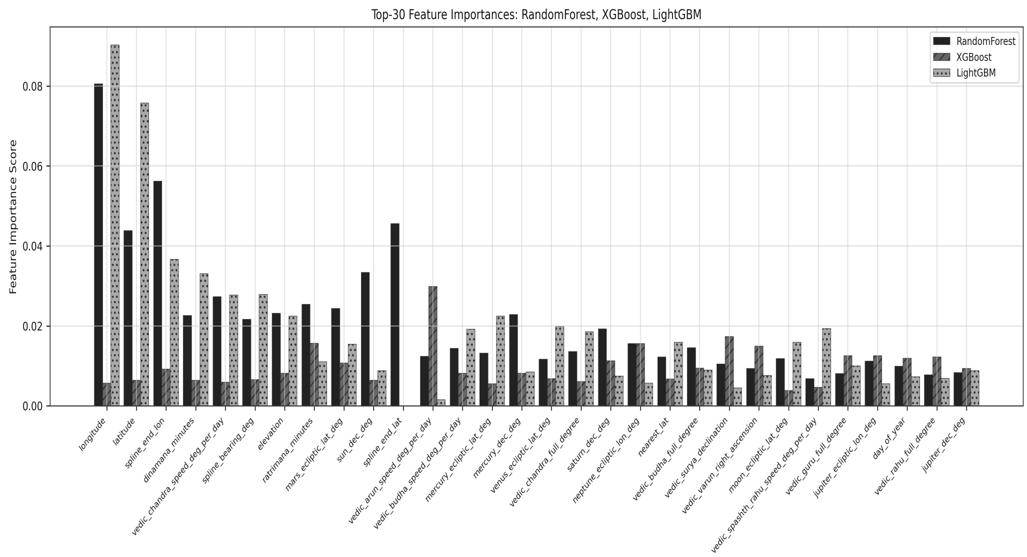

Fig 1. Top-30 feature importances (RF/XGB/LGB). Geographic features dominate.

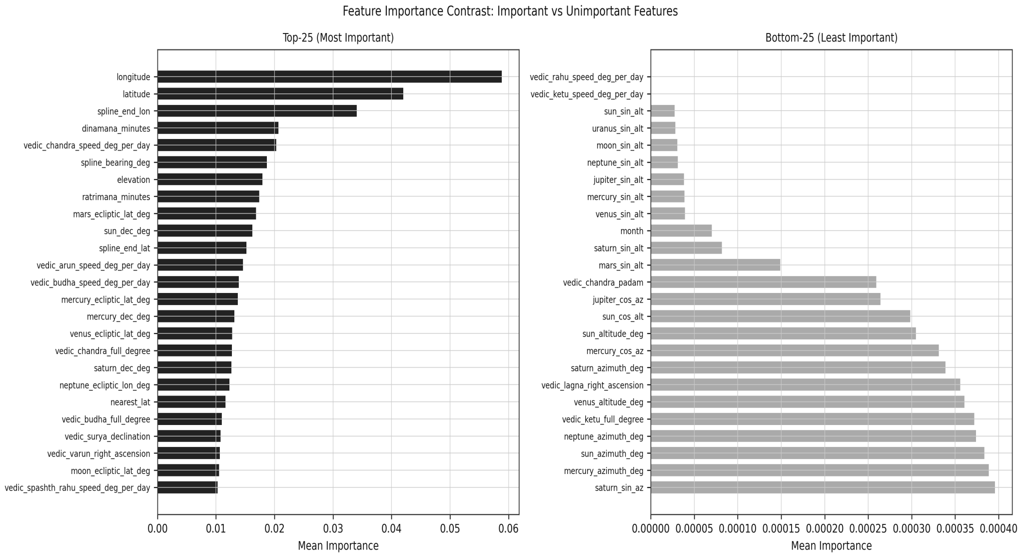

Fig 2. Top-25 vs bottom-25 mean importance contrast.

3. Mutual Information with Wind Speed

Mutual Information was computed using k-nearest-neighbour estimation (k=5) across all 186 features.

- lahiri_ayanamsha: MI = 0.720
- vedic_rahu_full_degree: MI = 0.691
- vedic_rahu_right_ascension: MI = 0.688
- Note: lahiri_ayanamsha and Rahu/Ketu degree features have high MI due to temporal trend confounding (precession), not genuine predictive signal.

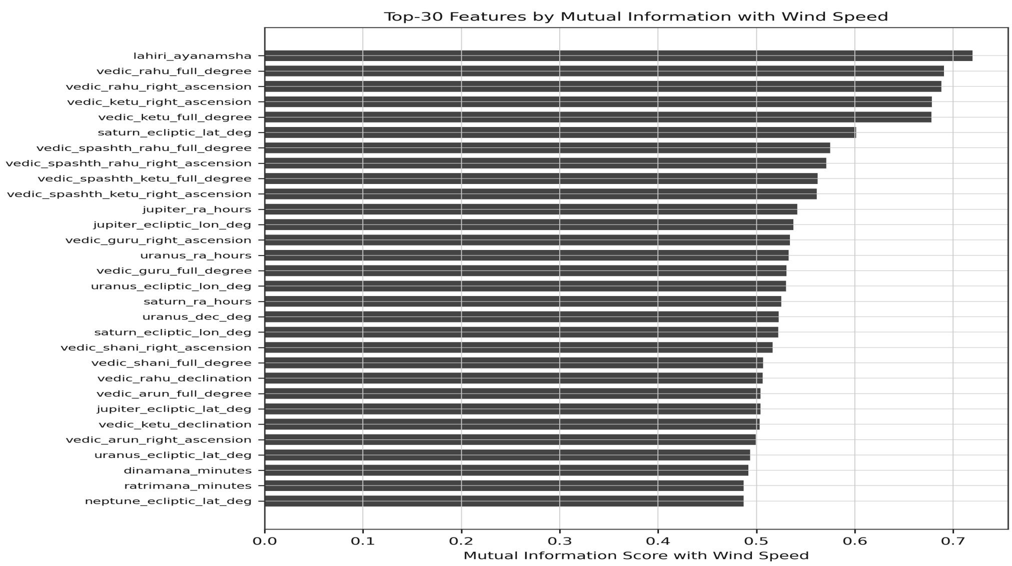

Fig 3. Top-30 Mutual Information scores with wind speed.

4. SHAP Explainability

SHAP TreeExplainer was applied to an XGBoost model. Mean |SHAP| values quantify average contribution magnitude per feature.

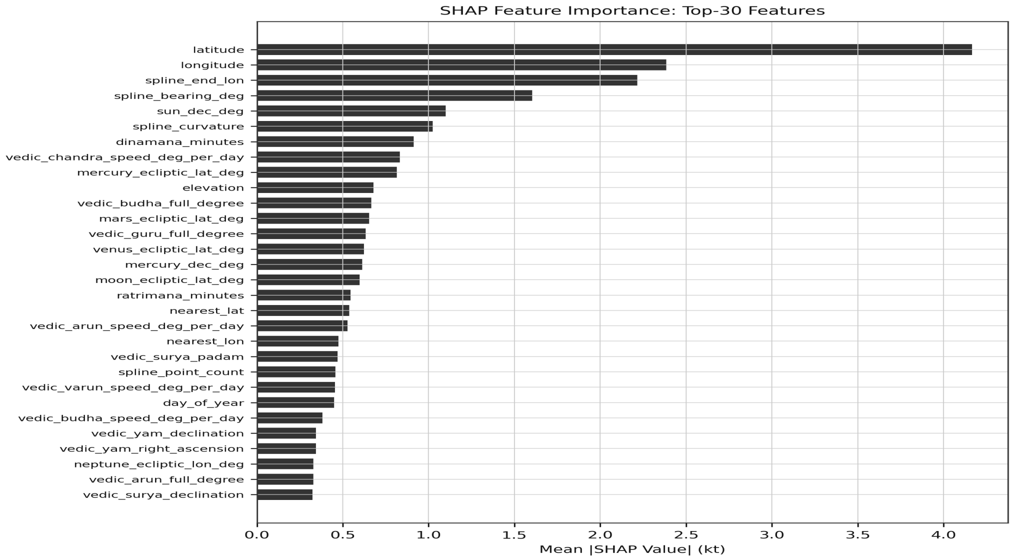

Fig 4. SHAP feature importance bar (top-30 mean |SHAP| values).

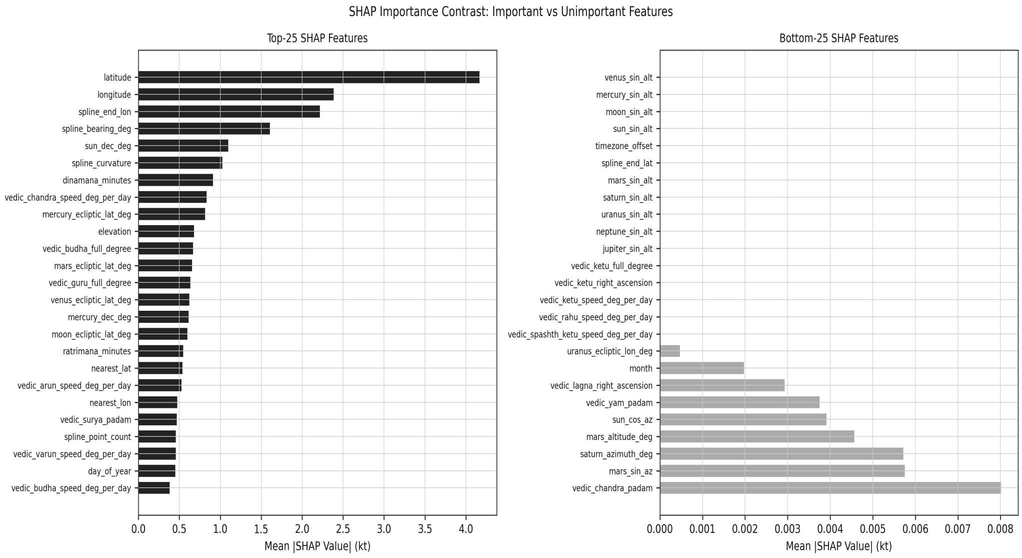

Fig 5. SHAP importance contrast: top-25 vs bottom-25.

4.1 SHAP Feature Interactions

Top-15 SHAP interaction pairs:

Feature 1	Feature 2	Mean Interaction
latitude	spline_bearing_deg	0.8220
latitude	spline_end_lon	0.5601
longitude	spline_end_lon	0.5514
longitude	latitude	0.5339
latitude	ratrimana_minutes	0.3631
latitude	vedic_varun_speed_deg_per_da	0.2816
latitude	vedic_arun_speed_deg_per_day	0.2572
latitude	sun_dec_deg	0.2566

latitude	vedic_budha_full_degree	0.2414
sun_dec_deg	mercury_dec_deg	0.2374
latitude	vedic_guru_full_degree	0.2361
vedic_budha_full_degree	dinamana_minutes	0.2217
latitude	dinamana_minutes	0.2174
latitude	vedic_chandra_right_ascensio	0.2067
sun_dec_deg	vedic_arun_speed_deg_per_day	0.1856

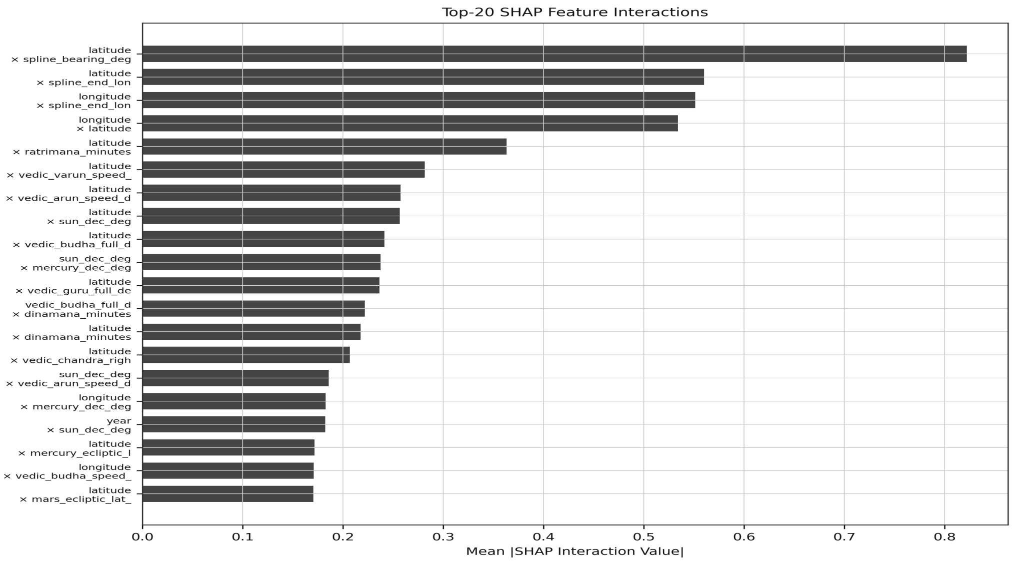

Fig 6. Top-20 SHAP feature interactions. latitude × spline_bearing_deg is dominant (0.822).

5. LIME and SHAP vs LIME Consensus

LIME was applied to 5 representative instances. 14 features appear in the top-30 of both SHAP and LIME (consensus set).

- Consensus features: dinamana_minutes, latitude, longitude, mercury_dec_deg, nearest_lat, neptune_ecliptic_lon_deg, ratrimana_minutes, spline_bearing_deg, spline_end_lon, sun_dec_deg.
- Vedic features in consensus: vedic_budha_speed_deg_per_day, vedic_budha_full_degree, vedic_guru_full_degree, vedic_arun_speed_deg_per_day.

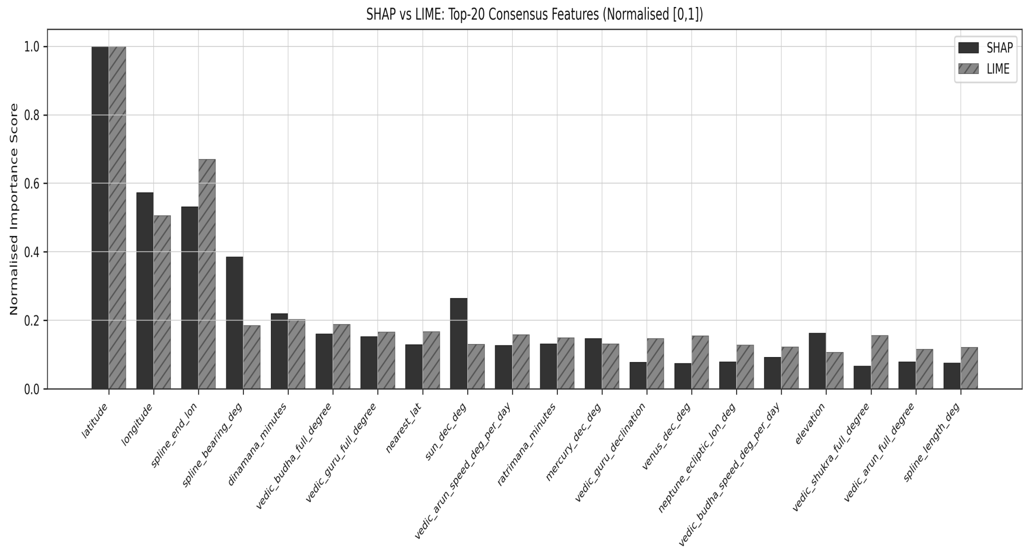

Fig 7. Normalised SHAP vs LIME importance for top-20 consensus features.

6. Vedic Category Analysis: Kruskal–Wallis Tests

Non-parametric Kruskal–Wallis tests compare wind speed across all Vedic categorical variables, with Benjamini–Hochberg FDR correction applied.

Variable	H	p-value	# Cat.	Sig.
drik_ritu	1214.03	2.68e-260	6	Yes
sunsign	1123.14	5.89e-234	12	Yes
yoga	143.35	3.43e-18	27	Yes
nakshatra	116.09	2.37e-13	27	Yes
drik_ayana	102.63	4.04e-24	2	Yes
moonsign	77.93	3.71e-12	12	Yes
tithi	51.26	7.50e-06	16	Yes
karana	42.48	6.15e-06	11	Yes
weekday	11.59	0.0717	7	No
paksha	8.11	0.0044	2	Yes

- drik_ritu (H=1214) and sunsign (H=1123) are calendar-season proxies, not independent Vedic signals.
- Large N (47,533) inflates all p-values. Effect sizes ($\varepsilon^2 < 0.003$) are negligible.

7. Nakshatra Intensity Analysis

Median wind speed was computed per Nakshatra. Kruskal–Wallis H=116, $p=2.4\times10^{\blacksquare 13}$, $\varepsilon^2\approx0.002$ (negligible effect size).

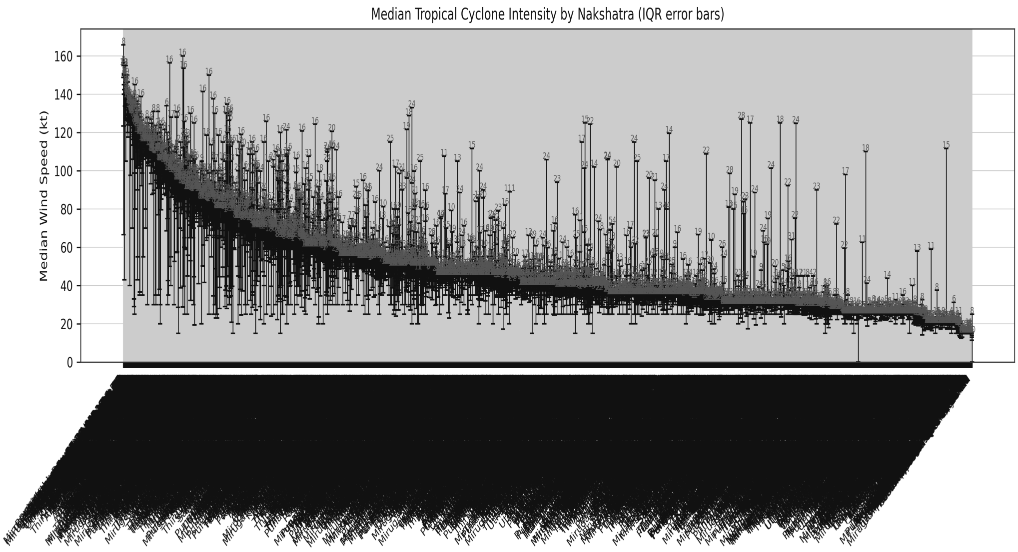

Fig 8. Median wind speed per Nakshatra with IQR error bars.

8. Tithi × Nakshatra Heatmaps

Cross-tabulation of Tithi (lunar day) × Nakshatra shows joint wind distribution. No systematic high-intensity zone is visible.

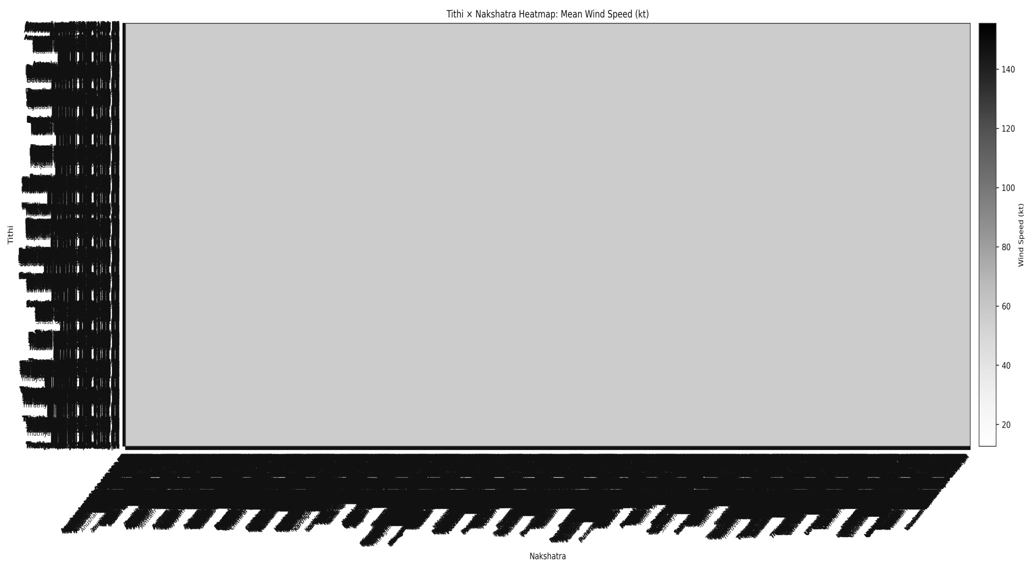

Fig 9. Mean wind speed (kt): Tithi × Nakshatra.

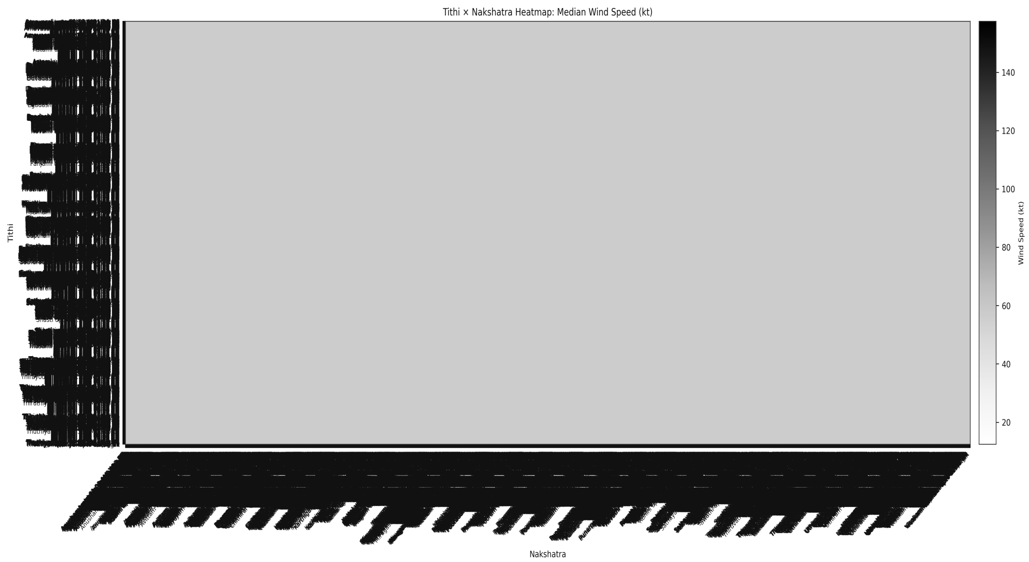

Fig 10. Median wind speed (kt): Tithi × Nakshatra.

9. Retrograde Planetary Motion and Storm Intensity

All 7 classical Vedic planets show statistically significant wind speed differences between retrograde and direct periods ($p < 0.001$), but effect sizes are negligible (rank-biserial $r < 0.10$ for all planets).

Planet	Retro%	Med(R)	Med(D)	∆Med	p-val	Sig.
guru	29.7%	40	42	-2.0	2.31e-23	Yes*
shani	46.3%	40	40	+0.0	4.99e-06	Yes*
mangal	9.1%	41	40	+1.0	0.0004	Yes*
budha	20.8%	40	40	+0.0	2.34e-22	Yes*
shukra	7.9%	45	40	+5.0	6.62e-06	Yes*
arun	51.8%	43	40	+3.0	1.02e-38	Yes*
varun	60.3%	43	39	+4.0	2.33e-59	Yes*

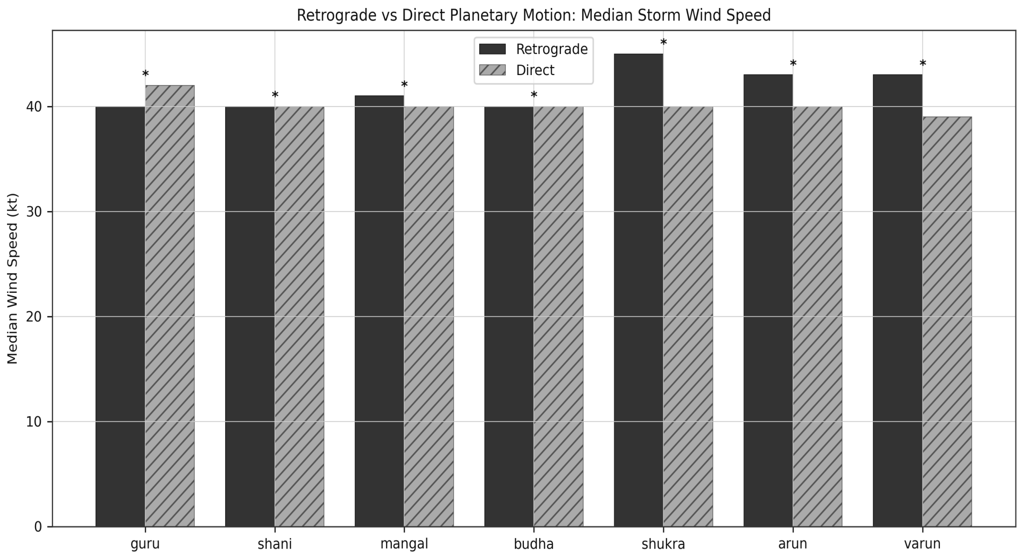

Fig 11. Retrograde vs direct median wind per planet. All differences ≤4 kt.

10. Planetary Aspect Correlations

Pearson correlations between planetary aspect angles and wind speed. Maximum |r| = 0.0766 (Jupiter–Saturn). All correlations are weak (|r| < 0.10).

Aspect Pair	Pearson r
aspect_jupiter_saturn	0.076636
aspect_moon_mars	0.033083
aspect_sun_saturn	0.028431
aspect_mercury_saturn	0.023740
aspect_moon_jupiter	0.015221
aspect_sun_mercury	0.014608
aspect_venus_saturn	0.013318
aspect_mercury_mars	0.007700
aspect_mars_saturn	0.004658
aspect_moon_saturn	0.004187
aspect_sun_mars	0.001534
aspect_moon_mercury	-0.000170
aspect_sun_moon	-0.000202
aspect_moon_venus	-0.002299
aspect_mercury_venus	-0.007633
aspect_mars_jupiter	-0.009821
aspect_sun_venus	-0.010910
aspect_mercury_jupiter	-0.016471
aspect_sun_jupiter	-0.017351
aspect_venus_mars	-0.025494
aspect_venus_jupiter	-0.039407

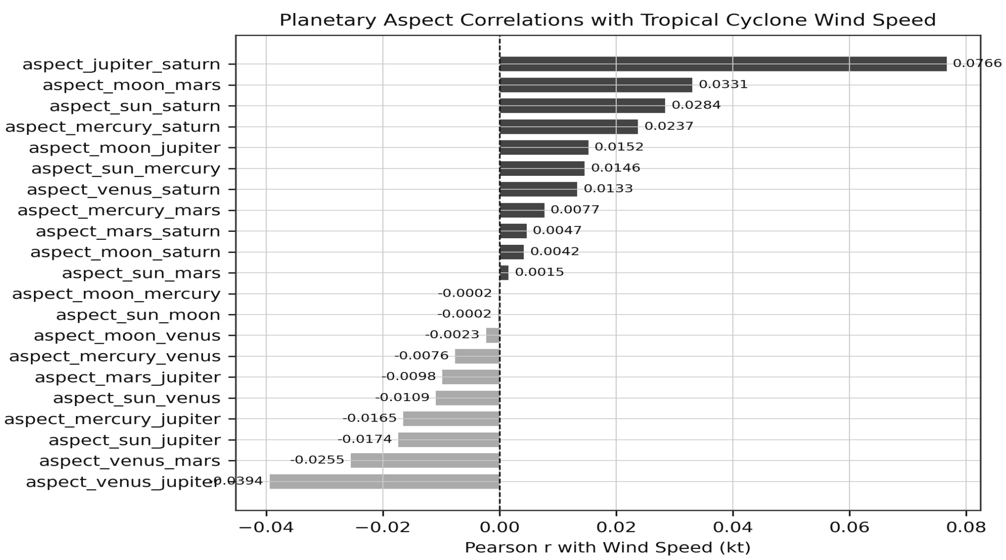

Fig 12. Planetary aspect correlations with wind speed.

11. Vedic Degree: Circular-Linear Correlations

Circular-linear method used (sin/cos decomposition of 0–360° degrees). Naive Pearson on raw degrees is incorrect for cyclical data.

Feature	r_sin	r_cos	r_circ
vedic_yam_full_degree	0.08938	-0.02018	0.08938
vedic_guru_full_degree	0.07130	-0.04249	0.07130
vedic_shani_full_degree	0.05394	-0.01491	0.05394
vedic_spashth_ketu_full_degree	0.02863	0.03457	0.03457
vedic_ketu_full_degree	0.02739	0.03392	0.03392
vedic_lagna_full_degree	-0.00098	0.00372	0.00372
vedic_chandra_full_degree	-0.00255	-0.00875	-0.00875
vedic_varun_full_degree	-0.00258	-0.00899	-0.00899
vedic_arun_full_degree	0.00522	-0.01990	-0.01990
vedic_rahu_full_degree	-0.02739	-0.03392	-0.03392
vedic_spashth_rahu_full_degree	-0.02863	-0.03457	-0.03457
vedic_mangal_full_degree	-0.00326	-0.05290	-0.05290
vedic_budha_full_degree	0.02773	-0.05700	-0.05700
vedic_surya_full_degree	0.04194	-0.06041	-0.06041
vedic_shukra_full_degree	0.02630	-0.06712	-0.06712

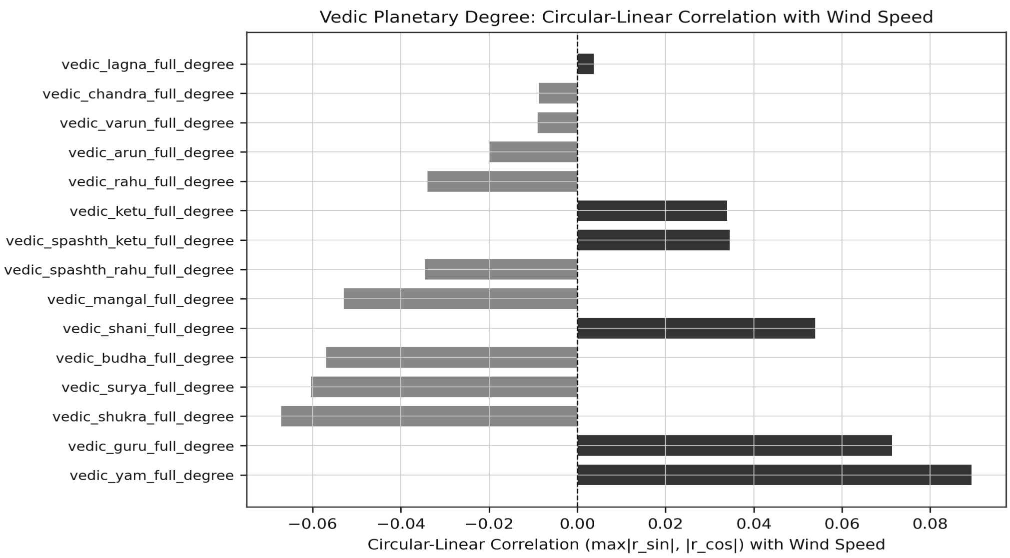

Fig 13. Circular-linear correlations per Vedic planet degree. Max r ≈ 0.089.

12. Variance Inflation Factor (VIF)

VIF analysis on 14 representative features. Thresholds: VIF < 5 = low, VIF 5–10 = moderate, VIF > 10 = high multicollinearity.

- Maximum VIF = 1.992 (feature: latitude).
- All values below 2.0 — negligible multicollinearity in this feature subset.

Feature	VIF	Assessment
latitude	1.992	Low
saturn_altitude_deg	1.709	Low

day_of_year	1.703	Low
jupiter_altitude_deg	1.579	Low
sun_ecliptic_lon_deg	1.481	Low
dinamana_minutes	1.471	Low
longitude	1.214	Low
sun_altitude_deg	1.096	Low
hour_utc	1.087	Low
elevation	1.079	Low
sun_azimuth_deg	1.029	Low
moon_ecliptic_lon_deg	1.010	Low
moon_altitude_deg	1.008	Low
moon_azimuth_deg	1.006	Low

13. Intensity Change Analysis

Wind change per 6-hour step. RI threshold: +30 kt/6h. RW threshold: –30 kt/6h.

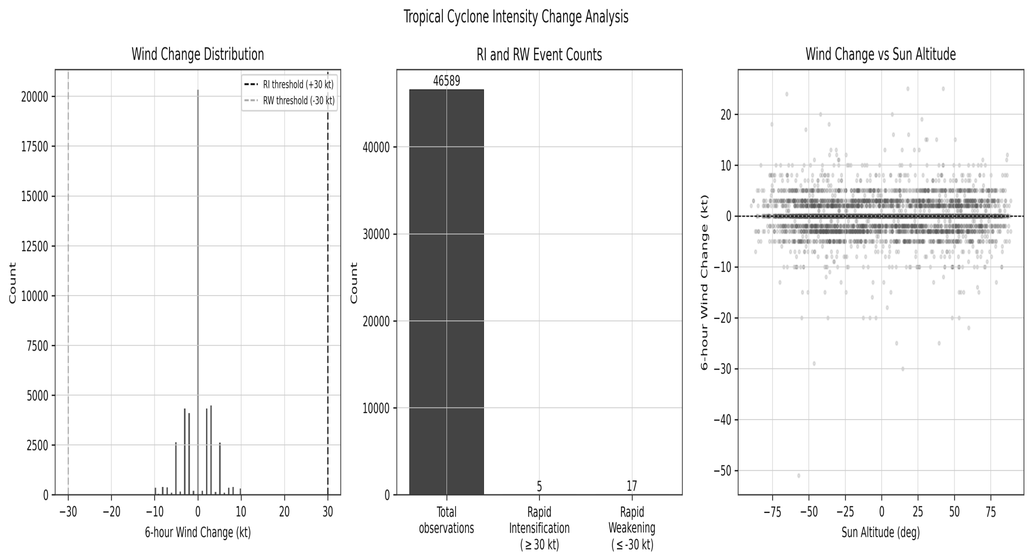

Fig 14. Wind change distribution, RI/RW event counts, and scatter vs sun altitude.

14. Conclusions

- Geographic features (latitude, longitude, spline geometry) dominate all importance metrics.
- Vedic categorical variables show statistically significant but negligible effect sizes ($\varepsilon^2 < 0.003$).
- All 7 retrograde planets are significant but with rank-biserial r < 0.10 (negligible practical effect).
- Circular-linear correlations of Vedic degrees: max r ≈ 0.089 (<1% variance explained).
- Planetary aspects: max |r| < 0.08 (weakest signals in dataset).
- SHAP–LIME consensus: 14 features; Vedic features in consensus: vedic_budha_speed_deg_per_day, vedic_budha_full_degree, vedic_guru_full_degree, vedic_arun_speed_deg_per_day.
- All VIF values below 2.0 for the analysed feature subset.